



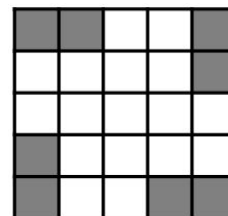
Checking understanding of percentages

Task A: Percentages, fractions and decimals

1) What proportion of the grid is shaded?

Give your answer as a percentage and a fraction.

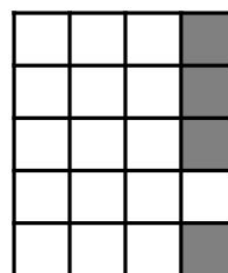
$$\frac{8}{25} = \frac{8 \times 4}{25 \times 4} = \frac{32}{100} = 32\%$$



2a) What proportion of the grid is shaded?

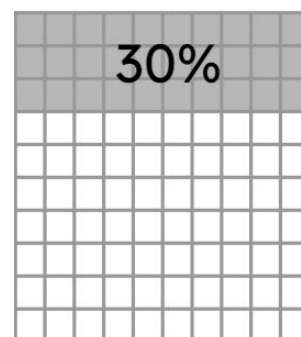
Give your answer as a percentage and a fraction.

$$\frac{4}{20} = \frac{4 \times 5}{20 \times 5} = \frac{20}{100} = 20\%$$

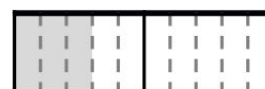


2b) What percentage is shaded on the 100 grid on the right?

Shade the same percentage on grids A to G



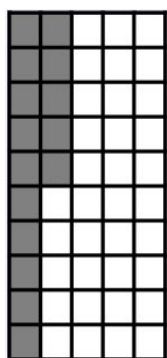
D



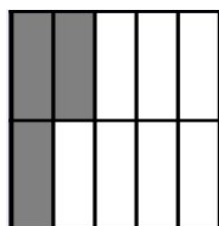
$$\begin{aligned} 30\% \text{ of } 2 &= \frac{30}{100} \times 2 \\ &= \frac{30 \times 2}{50 \times 2} = \frac{30}{50} \times \frac{2}{2} \\ &= \frac{3}{5} \times 1 = \frac{3}{5} \\ \frac{3}{5} \text{ of a block shaded} \end{aligned}$$

15 squares shaded

A



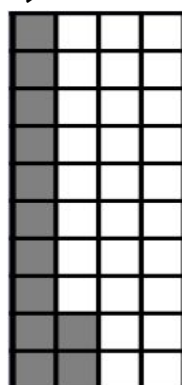
B



3 blocks shaded

12 squares shaded

C



Task B: Percentages of amounts

1) Work out the following:

a) 20% of 640

$$10\% \text{ of } 640 = 640 \div 10 = 64$$

$$20\% \text{ of } 640 = 2 \times 64 = 128$$

c) 15% of 8

$$10\% \text{ of } 8 = 8 \div 10 = 0.8$$

$$5\% \text{ of } 8 = 0.8 \div 2 = 0.4$$

$$15\% \text{ of } 8 = 0.8 + 0.4 = 1.2$$

b) 30% of 520

$$10\% \text{ of } 520 = 520 \div 10 = 52$$

$$30\% \text{ of } 520 = 3 \times 52 = 156$$

d) 35% of 30

$$10\% \text{ of } 30 = 30 \div 10 = 3$$

$$5\% \text{ of } 30 = 3 \div 2 = 1.5$$

$$35\% \text{ of } 30 = 3 \times 3 + 1.5 = 10.5$$

2a) Who do you think is correct? Why?



Aisha

Finding 10% of a number is easy. You just divide it by 10.

Finding 20% of a number is easy. You just divide it by 20.



Andeep

Finding 50% of a number is easy. You just divide it by 50.



Izzy

Aisha is correct. Andeep - 20% is found by dividing by five. Izzy - 50% would be divide by two.

You may also try to exemplify your reasoning using a particular value, showing why dividing it by 20 would not result in 20% or 50% of that number.

2b) What percentages would the incorrect calculation(s) actually find?

Izzy would have found 2%.

Andeep would have found 5%



3) Below are six students' different strategies for finding 75% of 250. One of them is incorrect.

Whose strategy is incorrect?

Explain the method used by each pupil.

Found 1% then multiplied by 75 to give 75%

Andeep

$$250 \div 100 = 2.5$$

$$2.5 \times 75 = ___$$

Aisha

$$250 \div 2 = 125$$

$$125 \div 2 = 62.5$$

$$125 + 62.5 = ___$$

Found 50% then found 25% and summed to make 75%

Found 10% then multiplied by 7 to give 70% then found 5% and summed to make 75%

Sofia

$$250 \div 10 = 25$$

$$25 \times 7 = 175$$

$$25 \div 2 = 12.5$$

$$175 + 12.5 = ___$$

Jacob

$$250 \div 4 = 62.5$$

$$250 - 62.5 = ___$$

Found 25% then subtracted from 100% to make 75%

Sam

$$250 \div 4 = 62.5$$

$$62.5 \times 3 = ___$$

Found 25% then multiplied by 3 to make 75%

Lucas is incorrect

Lucas

$$250 \div 75 = ___$$

